# The Bookmark and the Compass:
## Orientation Tools for Hypertext Users

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*The power of hypertext stems chiefly from the complex networks of conceptual links which hypertext documents can render explicit. This paper describes an array of orientation tools which help readers to choose a course through a complex information network without undue confusion or discomfort, while retaining the excitement and spontaneity users often experience when exploring complex information environments.*

A hypertext document engages its reader in a manner which a paper document cannot. Conventional media force the author to impose a single, permanent organization upon the document, while hypertext lets author and the reader cooperate to shape the document to the reader's personal inclinations. The reader becomes an active participant, the author's co-conspirator. The author, in turn, is freed from the necessity of adapting the document to the taste of an imagined "average" reader. Rather than write multiple documents intended to appeal to distinct audiences, the author can guide each reader through a series of choices which permit her to organize a personal path through the document.

It has been widely noted [Raskin 87] that, by offering the reader more choices and more opportunities for active participation, hypertext documents risk adding new sources of confusion to the reader's burdens [Conklin 87]. A person reading a linear document may, of course, lose the thread of the argument, but she may easily reorient herself by returning to the head of the paragraph or the start of the chapter. The hypertext reader, in contrast, may perceive herself to be enmeshed in a pathless network of information. It is desirable that the reader be able to choose what she will read next— indeed, it is the central premise of hypertext that this be so. But it would be regrettable indeed if this decision merely added to the reader's cognitive burdens the distracting question:

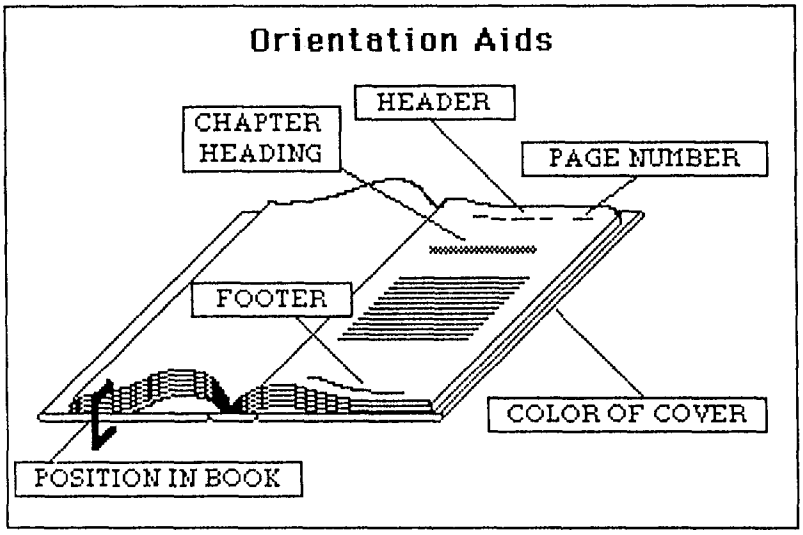
> *If this choice turns out to be a mistake, how can I get back?*

If her attention wanders or if a momentary interruption— perhaps a

telephone call— should occur, the hypertext user may find herself asking,

*Where am I? What am I reading?*

The physical design of familiar document forms provides frequent cues describing the reader's place in the document. Indeed, when we examine a paper book, a considerable fraction of our field of view is devoted to orientation cues.




These cues work together to give the reader a strong sense of context. If these cues are removed, as they are in many hypertext and online documentation systems, the reader may lose track of the context either through an external interruption or in the course of pursuing momentarily interesting links which prove to be dead ends. In either case, the reader suddenly realizes that the section of the document she is reading is not what she thought she wanted to read, and that no available action seems likely to advance her goals. Unable to form a plan, the reader feels lost, disoriented, and intensely frustrated.

Disorientation is always an unsettling experience. It can, however, be tolerated in some hypertext applications. College assignments [Yankelovich 87], for instance, are typically read with a single purpose in mind— review of a student's thoughts and assignment of a grade— and their reader is unlikely to become lost, even if briefly interrupted. Public information [Stallman 84] and library systems [Nelson 81] are sometimes designed to encourage spontaneous browsing; such systems do not try to reduce disorientation, but strive instead to reduce reader anxiety, to assure the reader that disorientation is normal and even desirable. Hypertext documents prepared for museum exhibits [Shneiderman and Morariu 86] may

reduce disorientation by chunking information and enforcing frequent returns to a "front page", thereby encouraging short sessions when many people are waiting in line.

The demands of technical manuals, of scientific and engineering documents, and of detailed business strategic documents, make it essential that disorientation be prevented whenever possible. Technical communicators have looked eagerly to the development of hypertext techniques [Thorsen and Bernstein 87], for hypertext offers an opportunity to make comprehensive and detailed instructions available to every reader, and yet does not require every reader to skim through an intimidating array of irrelevant information. Where current practice requires multiple documents, a single hypertext document could suffice. Indeed, because the audience rarely conforms to our archetypical images of "managers", "engineers", and similar conventional categories, unified hypertext might better satisfy all readers while obviating the expensive practice of producing a separate version for each audience.

These gains would be completely lost, however, if readers were prone to become disoriented; if the reader begins to fear that she is overlooking crucial information or if she feels lost in a maze of hypertext links, the reader will abandon hypertext and insist upon conventional media.

We know of no all-purpose tool that remedies the many sources of disorientation that hypertext readers encounter. Instead, we have chosen to explore a hypertext environment that offers many small tools to assure readers that they are not lost, and to help disoriented readers to find their way. These tools have been embedded in a new, implemented hypertext system, called Hypergate™ [Bernstein 88], which is intended for complex technical monographs and business documents. Many individual tools of the sort we describe have been used in previous hypertext systems, and we discuss our conscious borrowings below. We have, of course, been inspired and influenced by experiences with, and accounts of, hundreds of software systems, and offer our gratitude and apologies to the authors of systems from which we may have borrowed unconsciously.

The illustrations that follow are based on actual Hypergate™ displays. We have redrawn or retouched the figures to emphasize features under discussion and to remove extraneous details.

## Presentation, Layout, and Links

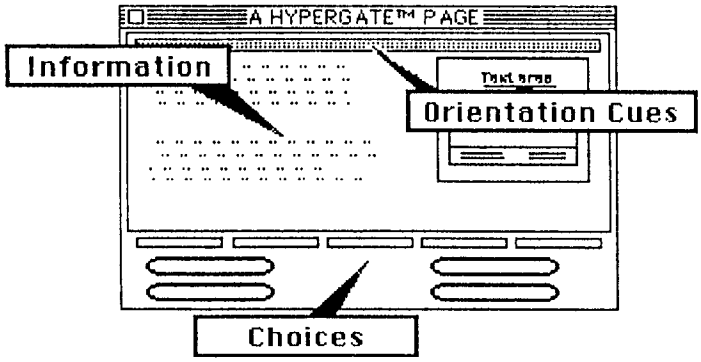
At every moment, hypertext systems provide three distinct kinds of

information:

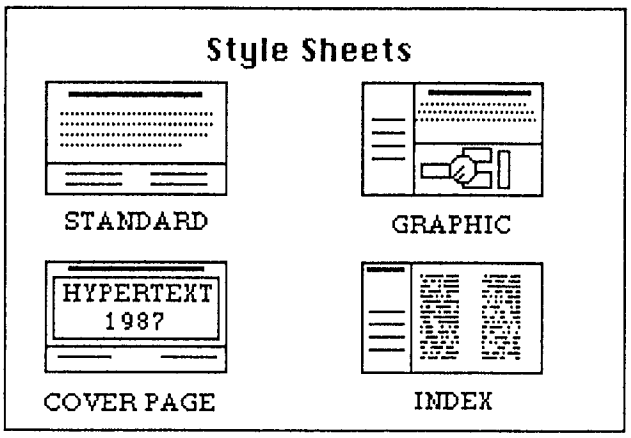
- Text, graphics, and other informational media
- Choices and links to additional information
- Orientation cues

As in any medium, the physical design of the hypertext page should reflect each distinct function. The exceptionally high information density of typical hypertext pages, on the other hand, poses a considerable challenge to the designer, a challenge. On the one hand, early hypertext systems have been criticized for busy and confusing displays. Some designers have sought to resolve the problem by planning complex display systems with stereoscopic graphics, sound, and tactile feedback [Lenat 83]. To avoid busy displays, some systems [Weyer 85] emulate the appearance of conventional linear documents; Guide™ [OWL 87] and InterText [Intermedia 85], for instance, display a scrolling linear text with subtle indications of the presence of hypertext links . Nelson [Nelson 81], similarly, envisions a side-by-side presentation of a pair of psuedo-linear documents with overlay graphics to display links between documents.

While these almost-linear hypertext systems have proved very attractive, we wished to explore hypertext systems which actively emphasized nonlinearity. We therefore divided the screen into discrete sections, and dedicated each sectionto one hypertext function.




The information area, in which text, graphics, animation sequences and simulations may be presented, is small. Because the information area does not scroll, there is no implicit linear order. The hypertext links that the author thinks are most important are prominently presented at the bottom of the page, while minor links (footnotes, minor arguments, technical details) are represented by small icons inside the text area. A narrow band at the top of the window presents orientation information, which the disoriented reader may supplement by pointing to the menu.




While Hypergate™ insists that the basic hypertext functions be represented by distinct screen areas, detailed layout is determined by a *style sheet.* The separation of different function helps readers orient themselves to new layouts, and the prominence of the choice buttons in all layouts ensures that the reader instantly knows where she might like to go next.

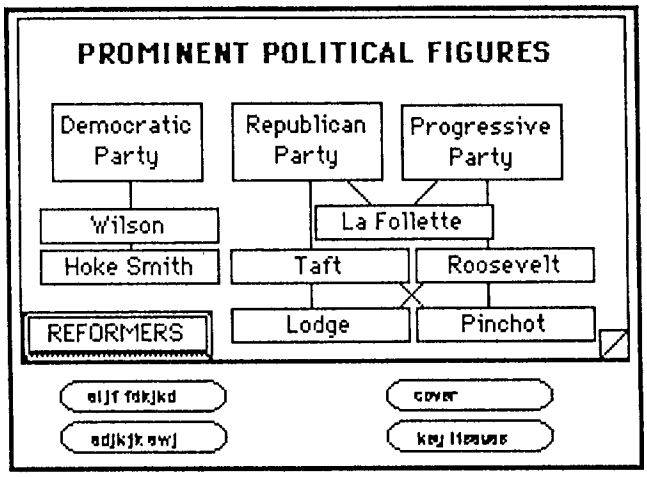
**Maps and Indexes**

Many hypertext provide maps or other special tools to help readers chart a course through the hypertext nextwork. In strictly heirarchical systems [Symbolics 86] and outline processors, a list of the parents, siblings, and children of the current node is a sufficient guide to local navigation. This information is easily conveyed textually, through easily-generated graphics, or through Smalltalk-style browsers [Halasz 87].

More complex networks suggest the need for larger and more complicated maps— maps that offer global information of general utility in addition to local orientation. Intermedia [Intermedia 85], for example, provides a window that displays all inter-document links in a "web" of hypertext documents. The Intermedia map window thus shows long-range information while suppressing local details. This theme has been carried even further in SemNet [Fairchild et al. 87], which displays the information network as a three-dimensional framework and which implements automatic clustering and a "fisheye" scheme for obscuring detail in remote information objects.

We believe it may be impossible to generate satisfactory and comprehensive maps without understanding the contents of the document. Idiomatic hypertext networks are frequently non-planar graphs, and we know of no presentation for such graphs which is aesthetically pleasing and which uses screen space efficiently. In a comprehensive map, many nodes must be suppressed to keep the

display from becoming a meaningless tangle, but choosing the nodes which are to be shown demands some understanding of the reader's goal and of the contents of individual nodes.




Rather than struggle to generate acceptable maps automatically, Hypergate™ shifts the burden of cartography to the author. Authors are encouraged to construct many maps, using their preferred graphic tools, and to paste them into their hypertext documents.

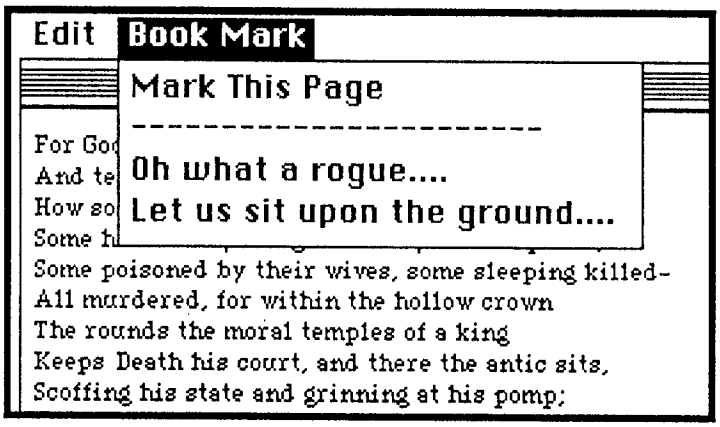
Manual cartography permits authors to include valuable rhetorical cues in the design of information maps. By graphically deemphasizing a massive technical reference section, Kaehler [Kaehler 1988] reassured new users of a complex software product that the product was easy to use. Careful use of layout and emphasis can draw the reader's eye to sections that the author considers especially important, while still ensuring that less-important sections remain easily accessible.

## Book Marks

Although careful organization, thoughtful presentation, and powerful navigational aids all help keep readers from becoming disoriented, they may not suffice to alleviate the reader's anxiety. *"I am not lost yet"*, the reader thinks, *"but this getting complicated. If I start wandering around, I may get lost."* In this way, fear of imminent disorientation might prevent readers from discovering and reading important sections of the document.

Hypertext book marks let the reader assuage this disorientation anxiety. If the reader begins to feel anxious, she simply marks her place in the book; should the results of a foray prove unsatisfactory or should she become lost, the book mark lets the reader retreat directly

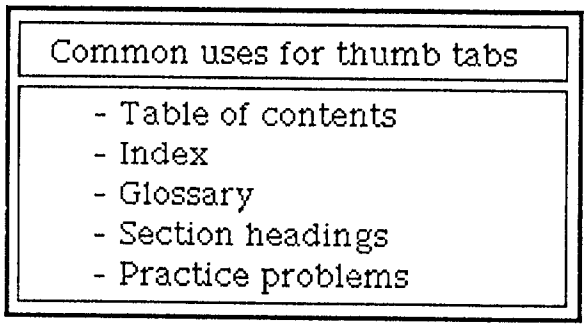
to a familiar location of her own choosing.




Book marks let the reader modify a complex and potentially intimidating document in a way that makes the document simpler and more personal. In our system, book marks are listed on a pull-down menu. Readers can mark or unmark any page they read, or can flip immediately to any marked page.

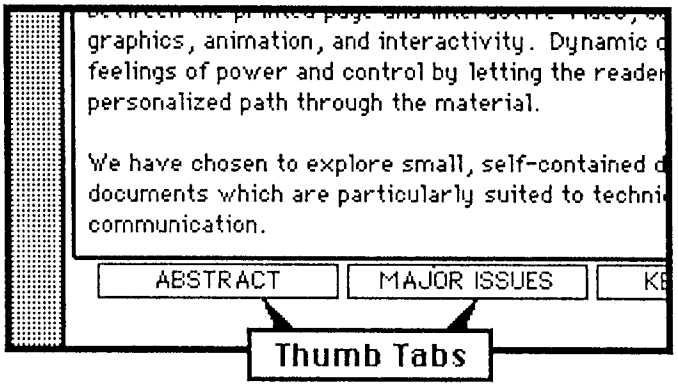
### Thumb Tabs

Book marks let readers mark parts of the document of personal significance. Thumb tabs, in turn, mark parts of the document that which the author knows will be useful to all readers. By pointing to a thumb tab on the display, the reader can jump directly to this useful information. Thumb tabs are useful for giving readers fast access to important parts of the document, both to sections that readers need to use often and to sections that readers can use to reorient themselves.




Thumb tabs are constant and unchanging; the same thumb tabs appear on every page of a document. Because thumb tabs never change and are always available for use, readers are assured that, should they be disoriented or follow an unproductive trail, they can

easily return to familiar ground.

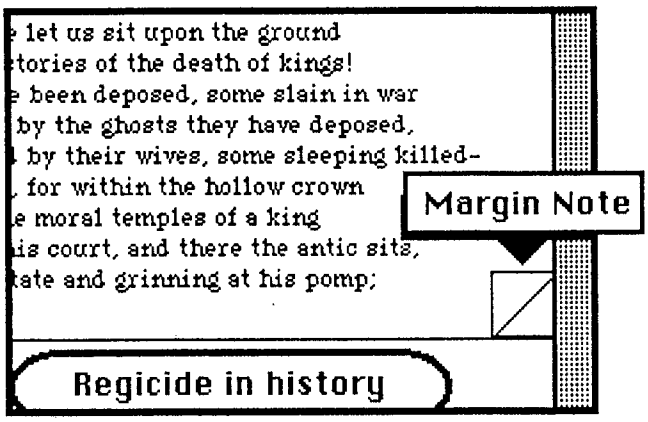



Book marks and thumb tabs find clear parallels in the way people learn to find their way in familiar cities [Lynch 60]. Thumb tabs are monuments and landmarks, places everyone knows— the Washington Monument or Hyde Park Corner. Bookmarks are familiar, personal places— the big tree in the zoo, or my bus stop. People use both kinds of landmarks to orient themselves in their own cities; hypertext documents should provide parallel facilities.

Many systems support thumb tabs, but often the facility is available only if the user takes a deliberate action. The Texas Instruments Explorer Glossary System [Texas Instruments 1985], for example, provides 26 thumb tabs in the window margin, labelled with each letter in the alphabet. INFO [Stallman 84] permits readers to jump to any page whose name they enter, a thumb tab facility which is always available though never displayed. Boxer [diSessa and Abelson 86] supports uniform commands to "pop" the user from her current page to a more general page; since Boxer documents are rarely nested very deeply, a few such "pops" should suffice to find the cover page. Multi-pane hypertext systems like NoteCards [Halasz 87] and Hypermedia [Garrett 86] encourage readers to leave previously-read pages open on the screen, thus permitting the prudent reader to relocate familiar islands at will. Nevertheless, both systems include implicit thumb tab commands that display a browser (NoteCards) or map (Intermedia) of nearby nodes.

## Margin Notes

Readers of paper documents sometimes write in the margins, underline sections of the text, crease significant page, or find other ways to physically modify the document. Others leave no mark at all upon the book— indeed, some readers express distaste for documents which others have marked. Hypertext can assist both audiences.




To those who take notes, hypertext offers powerful facilities for writing, revising, and reviewing their notes. Margin notes are written using a full text editor, and existing notes can be read (and edited) by pointing to the "note" icon. Margin notes cannot contain hypertext links to other pages, but they may— and often to— suggest useful paths or warn of possible pitfalls. Some writers use margin notes as footnotes, providing brief citations or pointing out exceptions to arguments in the main text.

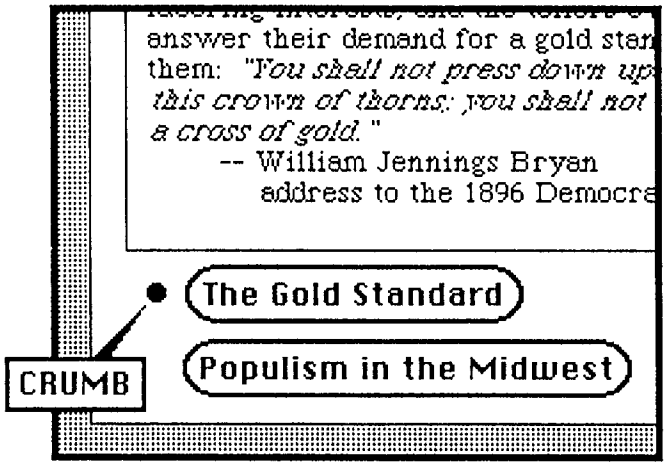
Note, however, that margin notes do not clutter the page or interfere with subsequent readers. Unless the reader asks to see the note, she sees only a small icon (and easily removed) designed to avoid excessive distraction.

## Breadcrumbs

Our discussion of orientation tools has emphasized tools that clarify the reader's options, that tell her where she can go next. Hypertext readers may also need help remembering where they have been.

- New information or sudden insight may lead a reader to review previously-read material.

- New pages may include links to material the reader has already seen. Knowing that she has already seen some items may influence the choice the reader makes.




We display a small marker— a bread crumb left on the trail as the reader passed this way earlier— to indicate choices which lead to material the reader has already seen. Crumbs help readers avoid undesired repetition. Just as important, crumbs guide disoriented readers to familiar territory which is near their current position. For example, crumbs will often mark references to introductory material and overviews, to which readers may refer many times as they peruse a complex document, or to maps, charts, or diagrams which have broad utility.

Upon request, the reader may ask for a menu of recently visited pages, and may jump directly to any such page. We have found it useful to limit the length of the list in order to avoid confronting the user with a huge menu. Thus, crumbs represent pages we have read *recently*, and imaginary birds remove bread crumbs the reader leaves unvisited for more than thirty pages.

## Summary

The promise of hypertext can only be realized if hypertext reading is both pleasant and efficient. Disorientation is unpleasant and wastes time; it must be avoided wherever possible. In a world of complex documents and frequent interruptions, however, we believe that occasional disorientation is inevitable, but that its worst effects can be remedied by orientation cues which are

- prominent (but not distracting)
- ever present
- predictable.

Automatic mapping and index-making tools are extremely desirable, but we know of no general method for adequately generating and

presenting this information. Instead, we encourage authors to build maps and indexes by hand, based on their understanding of the subject and of their audience. Tools to help authors build, revise, and test these complex pages will prove extremely useful, and have heretofore received scant attention. We believe that the solution to orienting readers within complex hypertext networks lies not in computer assisted mapping tools, but rather in well-crafted hand tools which will help authors to intelligently and sensitively instruct diverse audiences in the use of complex documents.

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